

The minimum weighted triple-minimum support on five labels

Research proof; literature priority not claimed

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Theorem 1. *On five or six labels, the smallest support of a probability distribution whose minimum is uniform on every triple is six. Arbitrary positive weights on the supporting permutations are allowed.*

Proof. The six-permutation family

012345, 521340, 531204, 413250, 423105, 032145

gives the upper bound on six labels, and deletion gives it on five. Its triple-minimum counts are verified in `verify_nine_dice_cloning.py`. The weighted support theorem in `weak_minimum_equality_obstruction.tex` gives the matching lower bound on six labels, and excludes support at most four on five labels. It remains to exclude a five-row distribution on five labels.

For every triple the five positive row weights are partitioned into three classes of total weight $1/3$. At least one class is a singleton, so at least one row has weight $1/3$. There can be at most two such rows. If there are two, they are singleton classes for every triple; the other three rows have the same minimum for every triple. Merge these three rows into any one of them with their combined weight $1/3$. This gives a forbidden three-row distribution on five labels.

Suppose there is exactly one row of weight $1/3$. In every triple, the remaining four rows split into two pairs of weight $1/3$. There are three perfect matchings on four rows. If at most two of these matchings have both pair weights $1/3$, there is a fixed two-element set of rows meeting every admissible pair once: for two matchings take a bipartition of their four-cycle; for one matching choose one endpoint of each pair. Together with the row of weight $1/3$, these two rows have three different minima on every triple. Giving the three selected rows equal weights creates another forbidden three-row distribution.

All three matchings must therefore be admissible. Their pair-sum equations force each of the four remaining row weights to be $1/6$. Duplicating the weight- $1/3$ row gives six equally weighted rows, two of which are identical. Relabel so that these two rows are 12345.

In every other row, the smallest label of any triple in the usual order can never be its minimum: the two identical rows have already used its entire quota. Consequently each other row starts with 4 or 5. The triple $\{3, 4, 5\}$ implies that exactly two rows start with each of these labels.

Fix $a < b \leq 3$. In the triple $\{a, b, 5\}$, the two rows starting with 5 use its quota, while the duplicated rows use a 's quota. Thus both rows starting with 4 must have $b < a$. Interchanging 4, 5 shows that both rows starting with 5 also have $b < a$. All four rows therefore restrict to 321 on $\{1, 2, 3\}$. The minimum of this triple is 3 four times and 2 never, a contradiction. $\square$

Together with the three-row construction on four labels, this gives weighted minimum support values 3, 3, 6, 6 for $n = 3, 4, 5, 6$ respectively. This concerns the minimum on triples only; it does not impose pairwise-unbiased comparisons.